Model Architecture: gemma-3-12b-it

Generated by sg_modelload

July 21, 2026

1 Model Overview

Model Name	gemma-3-12b-it
Path	/models/gemma-3-12b-it
Total Parameters	12,187,325,040
Weight Dtype	bfloat16

2 Forward Pass Algorithm

Algorithm 1 Forward pass of gemma-3-12b-it

$h \leftarrow \text{embed_tokens}(x) \in \mathbb{R}^{B \times T \times 1152}$
for $layer \leftarrow 0$ **to** 47 **do**
    $h \leftarrow \text{layer}_{layer}(h)$
end for
$h \leftarrow \text{norm}(h) \in \mathbb{R}^{B \times T \times 1152}$
$\text{logits} \leftarrow \text{lm_head}(h) \in \mathbb{R}^{B \times T \times 4096}$

3 Layer-by-Layer Architecture

All weight tensors and their dimensions are listed below.

Layer Path	Type	Weight Shape
model.language_model.layers.0	Gemma3DecoderLayer	–
model.language_model.layers.0.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.0.mlp	Gemma3MLP	–
model.language_model.layers.0.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.0.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.0.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.0.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.0.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.0.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.0.self_attn	Gemma3Attention	–
model.language_model.layers.0.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.0.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.0.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.0.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.0.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.0.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.0	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.0.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.0.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.0.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.0.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.0.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.0.mlp.fc1.weight	Linear	4304 × 1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.0.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.0.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.0.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.0.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.0.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.0.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.0.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.0.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.0.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.0.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.0.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.0.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.1	Gemma3DecoderLayer	–
model.language_model.layers.1.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.1.mlp	Gemma3MLP	–
model.language_model.layers.1.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.1.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.1.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.1.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.1.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.1.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.1.self_attn	Gemma3Attention	–
model.language_model.layers.1.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.1.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.1.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.1.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.1.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.1.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.1	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.1.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.1.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.1.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.1.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.1.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.1.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.1.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.1.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.1.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.1.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.1.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.1.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.1.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.1.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.1.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.1.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.1.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.1.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.2	Gemma3DecoderLayer	–
model.language_model.layers.2.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.2.mlp	Gemma3MLP	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.2.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.2.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.2.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.2.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.2.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.2.self_attn	Gemma3Attention	–
model.language_model.layers.2.self_attn.k_norm.weight	Gemma3RMSNorm	256

Layer Path	Type	Weight Shape
model.language_model.layers.2.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.2.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.2.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.2.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.2.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.2	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.2.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.2.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.2.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.2.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.2.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.2.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.2.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.2.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.2.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.2.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.2.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.2.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.2.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.2.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.2.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.2.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.2.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.2.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.3	Gemma3DecoderLayer	–
model.language_model.layers.3.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.3.mlp	Gemma3MLP	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.3.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.3.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.3.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.3.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.3.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.3.self_attn	Gemma3Attention	–
model.language_model.layers.3.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.3.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.3.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.3.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.3	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.3.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.3.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.3.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.3.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.3.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.3.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.3.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.3.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.3.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.3.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.3.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.3.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.3.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.3.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.3.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.3.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.3.self_attn.v_proj.weight	Linear	1152 × 1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.3.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.4	Gemma3DecoderLayer	–
model.language_model.layers.4.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.4.mlp	Gemma3MLP	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.4.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.4.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.4.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.4.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.4.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.4.self_attn	Gemma3Attention	–
model.language_model.layers.4.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.4.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.4.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.4.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.4.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.4.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.4	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.4.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.4.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.4.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.4.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.4.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.4.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.4.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.4.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.4.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.4.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.4.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.4.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.4.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.4.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.4.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.4.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.4.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.4.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.5	Gemma3DecoderLayer	–
model.language_model.layers.5.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.5.mlp	Gemma3MLP	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.5.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.5.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.5.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.5.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.5.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.5.self_attn	Gemma3Attention	–
model.language_model.layers.5.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.5.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.5.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.5.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.5.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.5.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.5	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.5.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.5.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.5.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.5.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.5.mlp	SiglipMLP	–

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.5.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.5.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.5.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.5.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.5.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.5.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.5.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.5.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.5.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.5.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.5.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.5.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.5.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.6	Gemma3DecoderLayer	–
model.language_model.layers.6.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.6.mlp	Gemma3MLP	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.6.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.6.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.6.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.6.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.6.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.6.self_attn	Gemma3Attention	–
model.language_model.layers.6.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.6.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.6.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.6.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.6.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.6.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.6	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.6.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.6.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.6.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.6.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.6.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.6.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.6.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.6.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.6.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.6.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.6.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.6.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.6.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.6.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.6.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.6.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.6.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.6.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.7	Gemma3DecoderLayer	–
model.language_model.layers.7.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.7.mlp	Gemma3MLP	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.7.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.7.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.7.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.7.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.7.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.7.self_attn	Gemma3Attention	–

Layer Path	Type	Weight Shape
model.language_model.layers.7.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.7.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.7.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.7.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.7	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.7.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.7.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.7.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.7.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.7.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.7.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.7.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.7.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.7.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.7.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.7.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.7.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.7.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.7.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.7.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.7.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.7.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.7.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.8	Gemma3DecoderLayer	–
model.language_model.layers.8.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.8.mlp	Gemma3MLP	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.8.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.8.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.8.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.8.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.8.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.8.self_attn	Gemma3Attention	–
model.language_model.layers.8.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.8.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.8.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.8.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.8.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.8.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.8	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.8.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.8.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.8.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.8.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.8.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.8.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.8.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.8.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.8.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.8.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.8.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.8.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.8.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.8.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.8.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.8.self_attn.q_proj.bias	Linear	1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.8.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.8.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.9	Gemma3DecoderLayer	–
model.language_model.layers.9.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.9.mlp	Gemma3MLP	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.9.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.9.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.9.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.9.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.9.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.9.self_attn	Gemma3Attention	–
model.language_model.layers.9.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.9.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.9.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.9.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.9.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.9.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.9	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.9.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.9.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.9.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.9.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.9.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.9.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.9.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.9.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.9.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.9.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.9.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.9.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.9.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.9.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.9.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.9.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.9.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.9.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.10	Gemma3DecoderLayer	–
model.language_model.layers.10.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.10.mlp	Gemma3MLP	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.10.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.10.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.10.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.10.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.10.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.10.self_attn	Gemma3Attention	–
model.language_model.layers.10.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.10.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.10.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.10.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.10.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.10.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.10	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.10.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.10.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.10.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.10.layer_norm2.bias	LayerNorm	1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.10.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.10.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.10.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.10.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.10.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.10.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.10.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.10.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.10.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.10.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.10.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.10.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.10.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.10.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.11	Gemma3DecoderLayer	–
model.language_model.layers.11.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.11.mlp	Gemma3MLP	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.11.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.11.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.11.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.11.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.11.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.11.self_attn	Gemma3Attention	–
model.language_model.layers.11.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.11.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.11.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.11.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.11.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.11	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.11.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.11.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.11.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.11.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.11.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.11.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.11.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.11.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.11.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.11.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.11.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.11.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.11.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.11.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.11.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.11.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.11.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.11.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.12	Gemma3DecoderLayer	–
model.language_model.layers.12.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.12.mlp	Gemma3MLP	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.12.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.12.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.12.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.12.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.12.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840

Layer Path	Type	Weight Shape
model.language_model.layers.12.self_attn	Gemma3Attention	–
model.language_model.layers.12.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.12.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.12.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.12.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.12.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.12.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.12	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.12.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.12.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.12.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.12.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.12.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.12.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.12.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.12.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.12.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.12.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.12.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.12.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.12.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.12.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.12.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.12.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.12.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.12.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.13	Gemma3DecoderLayer	–
model.language_model.layers.13.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.13.mlp	Gemma3MLP	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.13.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.13.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.13.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.13.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.13.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.13.self_attn	Gemma3Attention	–
model.language_model.layers.13.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.13.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.13.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.13.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.13.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.13.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.13	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.13.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.13.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.13.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.13.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.13.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.13.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.13.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.13.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.13.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.13.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.13.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.13.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.13.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.13.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.13.self_attn.q_proj.weight	Linear	1152 × 1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.13.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.13.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.13.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.14	Gemma3DecoderLayer	–
model.language_model.layers.14.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.14.mlp	Gemma3MLP	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.14.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.14.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.14.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.14.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.14.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.14.self_attn	Gemma3Attention	–
model.language_model.layers.14.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.14.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.14.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.14.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.14.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.14.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.14	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.14.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.14.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.14.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.14.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.14.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.14.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.14.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.14.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.14.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.14.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.14.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.14.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.14.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.14.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.14.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.14.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.14.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.14.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.15	Gemma3DecoderLayer	–
model.language_model.layers.15.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.15.mlp	Gemma3MLP	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.15.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.15.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.15.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.15.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.15.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.15.self_attn	Gemma3Attention	–
model.language_model.layers.15.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.15.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.15.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.15.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.15.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.15	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.15.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.15.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.15.layer_norm2.weight	LayerNorm	1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.15.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.15.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.15.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.15.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.15.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.15.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.15.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.15.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.15.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.15.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.15.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.15.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.15.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.15.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.15.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.16	Gemma3DecoderLayer	–
model.language_model.layers.16.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.16.mlp	Gemma3MLP	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.16.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.16.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.16.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.16.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.16.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.16.self_attn	Gemma3Attention	–
model.language_model.layers.16.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.16.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.16.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.16.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.16.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.16.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.16	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.16.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.16.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.16.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.16.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.16.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.16.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.16.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.16.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.16.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.16.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.16.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.16.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.16.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.16.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.16.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.16.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.16.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.16.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.17	Gemma3DecoderLayer	–
model.language_model.layers.17.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.17.mlp	Gemma3MLP	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.17.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.17.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.17.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.17.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840

Layer Path	Type	Weight Shape
model.language_model.layers.17.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.17.self_attn	Gemma3Attention	–
model.language_model.layers.17.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.17.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.17.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.17.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.17.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.17.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.17	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.17.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.17.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.17.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.17.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.17.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.17.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.17.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.17.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.17.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.17.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.17.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.17.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.17.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.17.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.17.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.17.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.17.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.17.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.18	Gemma3DecoderLayer	–
model.language_model.layers.18.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.18.mlp	Gemma3MLP	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.18.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.18.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.18.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.18.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.18.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.18.self_attn	Gemma3Attention	–
model.language_model.layers.18.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.18.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.18.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.18.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.18.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.18.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.18	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.18.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.18.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.18.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.18.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.18.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.18.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.18.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.18.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.18.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.18.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.18.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.18.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.18.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.18.self_attn.out_proj.bias	Linear	1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.18.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.18.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.18.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.18.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.19	Gemma3DecoderLayer	–
model.language_model.layers.19.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.19.mlp	Gemma3MLP	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.19.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.19.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.19.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.19.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.19.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.19.self_attn	Gemma3Attention	–
model.language_model.layers.19.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.19.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.19.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.19.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.19.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.19.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.19	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.19.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.19.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.19.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.19.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.19.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.19.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.19.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.19.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.19.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.19.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.19.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.19.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.19.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.19.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.19.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.19.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.19.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.19.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.20	Gemma3DecoderLayer	–
model.language_model.layers.20.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.20.mlp	Gemma3MLP	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.20.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.20.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.20.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.20.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.20.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.20.self_attn	Gemma3Attention	–
model.language_model.layers.20.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.20.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.20.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.20.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.20.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.20.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.20	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.20.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.20.layer_norm1.bias	LayerNorm	1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.20.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.20.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.20.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.20.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.20.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.20.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.20.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.20.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.20.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.20.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.20.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.20.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.20.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.20.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.20.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.20.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.21	Gemma3DecoderLayer	–
model.language_model.layers.21.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.21.mlp	Gemma3MLP	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.21.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.21.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.21.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.21.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.21.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.21.self_attn	Gemma3Attention	–
model.language_model.layers.21.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.21.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.21.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.21.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.21.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.21.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.21	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.21.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.21.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.21.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.21.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.21.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.21.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.21.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.21.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.21.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.21.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.21.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.21.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.21.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.21.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.21.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.21.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.21.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.21.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.22	Gemma3DecoderLayer	–
model.language_model.layers.22.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.22.mlp	Gemma3MLP	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.22.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.22.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.22.post_attention_layernorm.weight	Gemma3RMSNorm	3840

Layer Path	Type	Weight Shape
model.language_model.layers.22.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.22.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.22.self_attn	Gemma3Attention	–
model.language_model.layers.22.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.22.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.22.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.22.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.22.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.22.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.22	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.22.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.22.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.22.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.22.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.22.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.22.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.22.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.22.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.22.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.22.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.22.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.22.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.22.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.22.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.22.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.22.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.22.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.22.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.23	Gemma3DecoderLayer	–
model.language_model.layers.23.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.23.mlp	Gemma3MLP	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.23.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.23.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.23.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.23.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.23.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.23.self_attn	Gemma3Attention	–
model.language_model.layers.23.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.23.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.23.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.23.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.23.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.23	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.23.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.23.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.23.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.23.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.23.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.23.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.23.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.23.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.23.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.23.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.23.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.23.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.23.self_attn.out_proj.weight	Linear	1152 × 1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.23.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.23.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.23.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.23.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.23.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.24	Gemma3DecoderLayer	–
model.language_model.layers.24.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.24.mlp	Gemma3MLP	–
model.language_model.layers.24.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.24.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.24.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.24.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.24.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.24.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.24.self_attn	Gemma3Attention	–
model.language_model.layers.24.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.24.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.24.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.24.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.24.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.24.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.24	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.24.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.24.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.24.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.24.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.24.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.24.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.24.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.24.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.24.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.24.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.24.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.24.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.24.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.24.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.24.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.24.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.24.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.24.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.25	Gemma3DecoderLayer	–
model.language_model.layers.25.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.25.mlp	Gemma3MLP	–
model.language_model.layers.25.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.25.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.25.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.25.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.25.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.25.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.25.self_attn	Gemma3Attention	–
model.language_model.layers.25.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.25.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.25.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.25.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.25.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.25.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.25	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.25.layer_norm1.weight	LayerNorm	1152

Layer Path	Type	Weight Shape
model.vision_tower.encoder.layers.25.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.25.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.25.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.25.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.25.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.25.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.25.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.25.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.25.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.25.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.25.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.25.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.25.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.25.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.25.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.25.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.25.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.26	Gemma3DecoderLayer	–
model.language_model.layers.26.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.26.mlp	Gemma3MLP	–
model.language_model.layers.26.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.26.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.26.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.26.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.26.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.26.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.26.self_attn	Gemma3Attention	–
model.language_model.layers.26.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.26.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.26.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.26.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.26.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.26.self_attn.v_proj.weight	Linear	2048 × 3840
model.vision_tower.encoder.layers.26	SiglipEncoderLayer	–
model.vision_tower.encoder.layers.26.layer_norm1.weight	LayerNorm	1152
model.vision_tower.encoder.layers.26.layer_norm1.bias	LayerNorm	1152
model.vision_tower.encoder.layers.26.layer_norm2.weight	LayerNorm	1152
model.vision_tower.encoder.layers.26.layer_norm2.bias	LayerNorm	1152
model.vision_tower.encoder.layers.26.mlp	SiglipMLP	–
model.vision_tower.encoder.layers.26.mlp.fc1.weight	Linear	4304 × 1152
model.vision_tower.encoder.layers.26.mlp.fc1.bias	Linear	4304
model.vision_tower.encoder.layers.26.mlp.fc2.weight	Linear	1152 × 4304
model.vision_tower.encoder.layers.26.mlp.fc2.bias	Linear	1152
model.vision_tower.encoder.layers.26.self_attn	SiglipAttention	–
model.vision_tower.encoder.layers.26.self_attn.k_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.26.self_attn.k_proj.bias	Linear	1152
model.vision_tower.encoder.layers.26.self_attn.out_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.26.self_attn.out_proj.bias	Linear	1152
model.vision_tower.encoder.layers.26.self_attn.q_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.26.self_attn.q_proj.bias	Linear	1152
model.vision_tower.encoder.layers.26.self_attn.v_proj.weight	Linear	1152 × 1152
model.vision_tower.encoder.layers.26.self_attn.v_proj.bias	Linear	1152
model.language_model.layers.27	Gemma3DecoderLayer	–
model.language_model.layers.27.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.27.mlp	Gemma3MLP	–
model.language_model.layers.27.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.27.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.27.mlp.up_proj.weight	Linear	15360 × 3840

Layer Path	Type	Weight Shape
model.language_model.layers.27.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.27.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.27.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.27.self_attn	Gemma3Attention	–
model.language_model.layers.27.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.27.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.27.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.27.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.27.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.27.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.28	Gemma3DecoderLayer	–
model.language_model.layers.28.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.28.mlp	Gemma3MLP	–
model.language_model.layers.28.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.28.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.28.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.28.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.28.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.28.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.28.self_attn	Gemma3Attention	–
model.language_model.layers.28.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.28.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.28.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.28.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.28.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.28.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.29	Gemma3DecoderLayer	–
model.language_model.layers.29.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.29.mlp	Gemma3MLP	–
model.language_model.layers.29.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.29.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.29.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.29.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.29.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.29.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.29.self_attn	Gemma3Attention	–
model.language_model.layers.29.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.29.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.29.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.29.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.29.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.29.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.30	Gemma3DecoderLayer	–
model.language_model.layers.30.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.30.mlp	Gemma3MLP	–
model.language_model.layers.30.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.30.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.30.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.30.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.30.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.30.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.30.self_attn	Gemma3Attention	–
model.language_model.layers.30.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.30.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.30.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.30.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.30.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.30.self_attn.v_proj.weight	Linear	2048 × 3840

Layer Path	Type	Weight Shape
model.language_model.layers.31	Gemma3DecoderLayer	–
model.language_model.layers.31.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.31.mlp	Gemma3MLP	–
model.language_model.layers.31.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.31.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.31.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.31.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.31.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.31.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.31.self_attn	Gemma3Attention	–
model.language_model.layers.31.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.31.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.31.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.31.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.31.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.31.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.32	Gemma3DecoderLayer	–
model.language_model.layers.32.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.32.mlp	Gemma3MLP	–
model.language_model.layers.32.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.32.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.32.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.32.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.32.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.32.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.32.self_attn	Gemma3Attention	–
model.language_model.layers.32.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.32.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.32.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.32.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.32.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.32.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.33	Gemma3DecoderLayer	–
model.language_model.layers.33.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.33.mlp	Gemma3MLP	–
model.language_model.layers.33.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.33.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.33.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.33.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.33.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.33.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.33.self_attn	Gemma3Attention	–
model.language_model.layers.33.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.33.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.33.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.33.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.33.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.33.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.34	Gemma3DecoderLayer	–
model.language_model.layers.34.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.34.mlp	Gemma3MLP	–
model.language_model.layers.34.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.34.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.34.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.34.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.34.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.34.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.34.self_attn	Gemma3Attention	–

Layer Path	Type	Weight Shape
model.language_model.layers.34.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.34.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.34.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.34.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.34.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.34.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.35	Gemma3DecoderLayer	–
model.language_model.layers.35.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.35.mlp	Gemma3MLP	–
model.language_model.layers.35.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.35.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.35.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.35.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.35.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.35.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.35.self_attn	Gemma3Attention	–
model.language_model.layers.35.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.35.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.35.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.35.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.35.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.35.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.36	Gemma3DecoderLayer	–
model.language_model.layers.36.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.36.mlp	Gemma3MLP	–
model.language_model.layers.36.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.36.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.36.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.36.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.36.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.36.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.36.self_attn	Gemma3Attention	–
model.language_model.layers.36.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.36.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.36.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.36.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.36.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.36.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.37	Gemma3DecoderLayer	–
model.language_model.layers.37.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.37.mlp	Gemma3MLP	–
model.language_model.layers.37.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.37.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.37.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.37.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.37.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.37.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.37.self_attn	Gemma3Attention	–
model.language_model.layers.37.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.37.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.37.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.37.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.37.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.37.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.38	Gemma3DecoderLayer	–
model.language_model.layers.38.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.38.mlp	Gemma3MLP	–
model.language_model.layers.38.mlp.down_proj.weight	Linear	3840 × 15360

Layer Path	Type	Weight Shape
model.language_model.layers.38.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.38.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.38.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.38.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.38.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.38.self_attn	Gemma3Attention	–
model.language_model.layers.38.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.38.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.38.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.38.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.38.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.38.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.39	Gemma3DecoderLayer	–
model.language_model.layers.39.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.39.mlp	Gemma3MLP	–
model.language_model.layers.39.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.39.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.39.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.39.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.39.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.39.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.39.self_attn	Gemma3Attention	–
model.language_model.layers.39.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.39.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.39.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.39.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.39.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.39.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.40	Gemma3DecoderLayer	–
model.language_model.layers.40.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.40.mlp	Gemma3MLP	–
model.language_model.layers.40.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.40.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.40.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.40.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.40.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.40.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.40.self_attn	Gemma3Attention	–
model.language_model.layers.40.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.40.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.40.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.40.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.40.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.40.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.41	Gemma3DecoderLayer	–
model.language_model.layers.41.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.41.mlp	Gemma3MLP	–
model.language_model.layers.41.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.41.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.41.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.41.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.41.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.41.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.41.self_attn	Gemma3Attention	–
model.language_model.layers.41.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.41.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.41.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.41.self_attn.q_norm.weight	Gemma3RMSNorm	256

Layer Path	Type	Weight Shape
model.language_model.layers.41.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.41.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.42	Gemma3DecoderLayer	–
model.language_model.layers.42.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.42.mlp	Gemma3MLP	–
model.language_model.layers.42.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.42.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.42.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.42.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.42.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.42.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.42.self_attn	Gemma3Attention	–
model.language_model.layers.42.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.42.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.42.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.42.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.42.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.42.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.43	Gemma3DecoderLayer	–
model.language_model.layers.43.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.43.mlp	Gemma3MLP	–
model.language_model.layers.43.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.43.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.43.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.43.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.43.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.43.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.43.self_attn	Gemma3Attention	–
model.language_model.layers.43.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.43.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.43.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.43.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.43.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.43.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.44	Gemma3DecoderLayer	–
model.language_model.layers.44.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.44.mlp	Gemma3MLP	–
model.language_model.layers.44.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.44.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.44.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.44.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.44.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.44.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.44.self_attn	Gemma3Attention	–
model.language_model.layers.44.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.44.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.44.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.44.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.44.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.44.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.45	Gemma3DecoderLayer	–
model.language_model.layers.45.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.45.mlp	Gemma3MLP	–
model.language_model.layers.45.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.45.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.45.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.45.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.45.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840

Layer Path	Type	Weight Shape
model.language_model.layers.45.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.45.self_attn	Gemma3Attention	–
model.language_model.layers.45.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.45.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.45.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.45.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.45.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.45.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.46	Gemma3DecoderLayer	–
model.language_model.layers.46.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.46.mlp	Gemma3MLP	–
model.language_model.layers.46.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.46.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.46.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.46.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.46.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.46.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.46.self_attn	Gemma3Attention	–
model.language_model.layers.46.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.46.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.46.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.46.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.46.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.46.self_attn.v_proj.weight	Linear	2048 × 3840
model.language_model.layers.47	Gemma3DecoderLayer	–
model.language_model.layers.47.input_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.47.mlp	Gemma3MLP	–
model.language_model.layers.47.mlp.down_proj.weight	Linear	3840 × 15360
model.language_model.layers.47.mlp.gate_proj.weight	Linear	15360 × 3840
model.language_model.layers.47.mlp.up_proj.weight	Linear	15360 × 3840
model.language_model.layers.47.post_attention_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.47.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.47.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840
model.language_model.layers.47.self_attn	Gemma3Attention	–
model.language_model.layers.47.self_attn.k_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.47.self_attn.k_proj.weight	Linear	2048 × 3840
model.language_model.layers.47.self_attn.o_proj.weight	Linear	3840 × 4096
model.language_model.layers.47.self_attn.q_norm.weight	Gemma3RMSNorm	256
model.language_model.layers.47.self_attn.q_proj.weight	Linear	4096 × 3840
model.language_model.layers.47.self_attn.v_proj.weight	Linear	2048 × 3840
lm_head.weight	Linear	262208 × 384
model	Gemma3Model	–
model.language_model	Gemma3TextModel	–
model.language_model.embed_tokens.weight	Gemma3TextScaledWordEmbedding	262208 × 384
model.language_model.layers	ModuleList	–
model.language_model.norm.weight	Gemma3RMSNorm	3840
model.multi_modal_projector.mm_input_projection_weight	Gemma3MultiModalProjector	1152 × 3840
model.multi_modal_projector.mm_soft_emb_norm.weight	Gemma3RMSNorm	1152
model.vision_tower	SiglipVisionModel	–
model.vision_tower.embeddings	SiglipVisionEmbeddings	–
model.vision_tower.embeddings.patch_embedding.weight	Conv2d	1152 × 3 × 14
model.vision_tower.embeddings.patch_embedding.bias	Conv2d	1152
model.vision_tower.embeddings.position_embedding.weight	Embedding	4096 × 1152
model.vision_tower.encoder	SiglipEncoder	–
model.vision_tower.encoder.layers	ModuleList	–
model.vision_tower.post_layernorm.weight	LayerNorm	1152
model.vision_tower.post_layernorm.bias	LayerNorm	1152

4 Parameter Count Estimation

Component	Parameters	% of Total
MLP	8,761,356,144	71.9%
Attention	2,408,400,384	19.8%
Embedding	1,006,878,720	8.3%
LM Head	1,006,878,720	8.3%
Other	9,820,800	0.1%
LayerNorm	868,992	0.0%
Total	**12,187,325,040**	**100.0%**

4.1 Per-Layer Parameter Count

Layer	Params	Running Total
model.language_model.layers.0.input_layernorm	3,840	3,840
model.language_model.layers.0.mlp.down_proj	58,982,400	58,986,240
model.language_model.layers.0.mlp.gate_proj	58,982,400	117,968,640
model.language_model.layers.0.mlp.up_proj	58,982,400	176,951,040
model.language_model.layers.0.post_attention_layernorm	3,840	176,954,880
model.language_model.layers.0.post_feedforward_layernorm	3,840	176,958,720
model.language_model.layers.0.pre_feedforward_layernorm	3,840	176,962,560
model.language_model.layers.0.self_attn.k_norm	256	176,962,816
model.language_model.layers.0.self_attn.k_proj	7,864,320	184,827,136
model.language_model.layers.0.self_attn.o_proj	15,728,640	200,555,776
model.language_model.layers.0.self_attn.q_norm	256	200,556,032
model.language_model.layers.0.self_attn.q_proj	15,728,640	216,284,672
model.language_model.layers.0.self_attn.v_proj	7,864,320	224,148,992
model.vision_tower.encoder.layers.0.layer_norm1	2,304	224,151,296
model.vision_tower.encoder.layers.0.layer_norm2	2,304	224,153,600
model.vision_tower.encoder.layers.0.mlp.fc1	4,962,512	229,116,112
model.vision_tower.encoder.layers.0.mlp.fc2	4,959,360	234,075,472
model.vision_tower.encoder.layers.0.self_attn.k_proj	1,328,256	235,403,728
model.vision_tower.encoder.layers.0.self_attn.out_proj	1,328,256	236,731,984
model.vision_tower.encoder.layers.0.self_attn.q_proj	1,328,256	238,060,240
model.vision_tower.encoder.layers.0.self_attn.v_proj	1,328,256	239,388,496
model.language_model.layers.1.input_layernorm	3,840	239,392,336
model.language_model.layers.1.mlp.down_proj	58,982,400	298,374,736
model.language_model.layers.1.mlp.gate_proj	58,982,400	357,357,136
model.language_model.layers.1.mlp.up_proj	58,982,400	416,339,536
model.language_model.layers.1.post_attention_layernorm	3,840	416,343,376
model.language_model.layers.1.post_feedforward_layernorm	3,840	416,347,216
model.language_model.layers.1.pre_feedforward_layernorm	3,840	416,351,056
model.language_model.layers.1.self_attn.k_norm	256	416,351,312
model.language_model.layers.1.self_attn.k_proj	7,864,320	424,215,632
model.language_model.layers.1.self_attn.o_proj	15,728,640	439,944,272
model.language_model.layers.1.self_attn.q_norm	256	439,944,528
model.language_model.layers.1.self_attn.q_proj	15,728,640	455,673,168
model.language_model.layers.1.self_attn.v_proj	7,864,320	463,537,488
model.vision_tower.encoder.layers.1.layer_norm1	2,304	463,539,792
model.vision_tower.encoder.layers.1.layer_norm2	2,304	463,542,096
model.vision_tower.encoder.layers.1.mlp.fc1	4,962,512	468,504,608
model.vision_tower.encoder.layers.1.mlp.fc2	4,959,360	473,463,968
model.vision_tower.encoder.layers.1.self_attn.k_proj	1,328,256	474,792,224
model.vision_tower.encoder.layers.1.self_attn.out_proj	1,328,256	476,120,480
model.vision_tower.encoder.layers.1.self_attn.q_proj	1,328,256	477,448,736
model.vision_tower.encoder.layers.1.self_attn.v_proj	1,328,256	478,776,992
model.language_model.layers.2.input_layernorm	3,840	478,780,832
model.language_model.layers.2.mlp.down_proj	58,982,400	537,763,232
model.language_model.layers.2.mlp.gate_proj	58,982,400	596,745,632
model.language_model.layers.2.mlp.up_proj	58,982,400	655,728,032
model.language_model.layers.2.post_attention_layernorm	3,840	655,731,872
model.language_model.layers.2.post_feedforward_layernorm	3,840	655,735,712
model.language_model.layers.2.pre_feedforward_layernorm	3,840	655,739,552
model.language_model.layers.2.self_attn.k_norm	256	655,739,808
model.language_model.layers.2.self_attn.k_proj	7,864,320	663,604,128
model.language_model.layers.2.self_attn.o_proj	15,728,640	679,332,768
model.language_model.layers.2.self_attn.q_norm	256	679,333,024
model.language_model.layers.2.self_attn.q_proj	15,728,640	695,061,664
model.language_model.layers.2.self_attn.v_proj	7,864,320	702,925,984
model.vision_tower.encoder.layers.2.layer_norm1	2,304	702,928,288
model.vision_tower.encoder.layers.2.layer_norm2	2,304	702,930,592
model.vision_tower.encoder.layers.2.mlp.fc1	4,962,512	707,893,104
model.vision_tower.encoder.layers.2.mlp.fc2	4,959,360	712,852,464
model.vision_tower.encoder.layers.2.self_attn.k_proj	1,328,256	714,180,720
model.vision_tower.encoder.layers.2.self_attn.out_proj	1,328,256	715,508,976
model.vision_tower.encoder.layers.2.self_attn.q_proj	1,328,256	716,837,232